Lesson Plan: Comparative Molecular Modeling — From 2D Paper to 3D Electrostatic Potential

Grade Level: 9–11 (Chemistry / Honors Chemistry)

Target Molecules: Methane (CH_4), Fluoromethane (CH_3F), simple molecules students build themselves, and more complex structures imported in WebMO.

NGSS Alignment

- **Performance Expectations:**
  - **HS-PS1-1:** Use the periodic table as a model to predict the relative properties of elements based on patterns of electrons in the outermost energy level of atoms.
  - **HS-PS1-3:** Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.
  - **HS-PS2-4:** Use mathematical representations of Newton's Law of Gravitation and Coulomb's Law to describe and predict the gravitational and electrostatic forces between objects. *(Applied here to atomic radii and bond lengths)*.
- **Science & Engineering Practices (SEP):**
  - *Developing and Using Models:* Evaluate limitations and strengths of multiple models representing the same physical system.
  - *Analyzing and Interpreting Data:* Analyze computational visualizations to explain charge distribution and bond characteristics.
- **Disciplinary Core Ideas (DCI):**
  - *PS1.A: Structure and Properties of Matter:* The structure and interactions of matter at the bulk scale are determined by electrical forces within and between atoms.
- **Crosscutting Concepts (CCC):**
  - *Structure and Function:* The 3D geometry and electron density distribution within a molecule govern its physical properties.
  - *Systems and System Models:* Models are limited representations that isolate specific features of complex atomic systems.

Prerequisites & Prior Knowledge

- Periodic trends (atomic radii, electronegativity).
- Basic construction of 2D Lewis structures.
- Familiarity with Coulomb's Law $F = k \frac{q_1 q_2}{r^2}$ from prior units.

Learning Objectives

1. Construct and compare 2D Lewis structures, physical 3D MolyMod representations, and computational WebMO models for CH_4 and CH_3F (as well as additional student-generated WebMO Models).
2. Identify the limitations of physical model kits regarding atomic size differences and bond lengths.
3. Apply Coulomb's Law to explain how atomic radii influence bond length and attractive forces between nuclei and shared electrons.
4. Interpret WebMO Electrostatic Potential (ESP) maps to visualize how differences in electronegativity distort charge distribution in CH_3F compared to CH_4 (as well as additional student-generated WebMO Models).

Materials & Software

- Student whiteboards or lab notebooks.
- MolyMod Molecular Model Kits.
- Computer or Chromebooks with access to WebMO site
- Student Handout: "Comparing Molecular Models & Charge Distribution Worksheet".

Instructional Sequence

Phase 1: 2D Lewis Structures & The "Disney Cinderella Castle" Metaphor

1. **Task:** Students draw 2D Lewis structures for Methane (CH_4) and Fluoromethane (CH_3F) on their whiteboards.
2. **Pedagogical Framing (Teacher Metaphor):**
    - Address the limitation of flat drawings without introducing dash-and-wedge notation:
    - *"If you take a photo in front of Cinderella's Castle at Disney World, the photo itself is 2-dimensional, but you know it represents a snapshot of a real, 3-dimensional space. A Lewis structure on paper works the exact same way—it is a 2D snapshot that helps us track electrons, but the real molecule occupies 3D space."*
3. **Formative Check:** Verify that students correctly account for all valence electrons (8 in CH_4, 14 in CH_3F, including non-bonding pairs on Fluorine).

Phase 2: Physical 3D Modeling with MolyMods

1. **Task:** Students construct physical models of CH_4 and CH_3F using their MolyMod kits.
2. **Student Sense-Making:**
    - Students compare their physical models to their flat whiteboard drawings.
    - Formative feedback guides students to realize that electron domain repulsion forces the atoms into a tetrahedral arrangement in 3D space rather than a flat criss-cross.
3. **Target Observation:** Students note that in the MolyMod set, all bonds use identical plastic connectors, making C—H and C—F bonds look physically identical in length.

Phase 3: Transition to WebMO & Model Limitation Discussion

1. **WebMO Build Procedure:**
    - Open WebMO and create two separate tabs: Tab 1 for CH_4, Tab 2 for CH_3F.
    - Draw the rough atomic connections. Note: The initial build will look distorted ("worse for wear").
    - Navigate to: Cleanup → Geometry.
    - Observe how the software auto-adjusts the molecule into a symmetrical 3D shape similar to their MolyMod model.
2. **Formative Discussion (MolyMod Limitations):**
    - **Prompt:** *"Compare your MolyMod model to the optimized WebMO model. What does the MolyMod kit fail to tell us about the differences between Hydrogen and Fluorine?"*
    - **Target Realization:** MolyMods use uniform stick lengths and static sphere sizes. They do not account for relative atomic sizes or differences in actual bond lengths. They also do not represent the valence electrons of atoms within the molecular structure.

Phase 4: Connecting Coulomb's Law to Chemical Bonds

1. **Revisiting Coulomb's Law:**
    - Prompt students to recall Coulomb's Law $F = k \frac{q_1 q_2}{r^2}$: As distance (r) increases, attractive force decreases.
2. **Applying to Atomic Structure & Bonding:**
    - Guide a short class discussion on internal atomic forces: Protons in the nucleus attract an atom's own valence electrons, but in a covalent bond, the protons of Atom A also attract the valence electrons of Atom B.
    - **Key Concept:** Larger atoms have larger atomic radii, resulting in a greater internuclear distance (r) when bonded. This leads to longer, weaker bond interactions compared to smaller atoms with shorter bond distances.

Phase 5: WebMO Electrostatic Potential (ESP) Maps

1. **WebMO ESP Computation:**
    - In WebMO, open the optimized models for CH_4 and CH_3F.
    - Navigate to: Calculate → Huckel → Electrostatic Potential.
    - Render the ESP surface for both molecules.
    - We need to compare the ESP on the same scale.
        i. Options → Mapped Properties
        ii. Change Min: -0.17 Max: 0.13
2. **Analysis & Comparison:**
    - **Methane CH_4:** Students observe a uniform, non-shifting color gradient across the molecule, indicating an equal distribution of electron density.
    - **Fluoromethane (CH_3F):** Students observe a dramatic shift in color density (a distinct electron-rich region localized around the Fluorine atom due to its higher electronegativity).

Phase 6: WebMO Exploration:

1. Students can utilize WebMO to examine a variety of molecular structures.
    a. By employing diverse kinds of atoms and multiple chemical bonds, they can construct the molecular models independently. Water should be highly encouraged to build and explore.
    b. They can automatically generate recognized compounds, such as acetaminophen or aspirin, by searching via the Lookup → Name option.
    c. Interested students have the opportunity to conduct a more detailed investigation into the advanced features offered by WebMO.

Teacher Eyes Only: Model Comparison Matrix

Feature/Dimension	Model 1: Lewis Structure	Model 2: 3D MolyMod Kit	Model 3: 3D WebMO Simulation (ESP Map)
Primary Focus	Electron counting & atom connectivity	Tactile 3D spatial geometry	Calculated bond properties & charge distribution
Key Strengths	Quick to draw; clearly shows bonding vs non-bonding valence electrons	Physical representation of VSEPR shape and 3D spatial occupancy	Displays realistic bond distances, atomic size differences, and electron density distributions
Major Limitations	Distorts 3D space into a flat 2D plane	Uses fixed stick lengths and uniform sphere sizes; hides differences in bond length/atomic radii	Abstract computational interface
Coulombic Insights	None	Indirect (shows spatial arrangement only)	Directly maps how electronegativity and distance distort electrostatic forces across the surface.

Student Lab Handout

Lab Activity: Comparison of Three Different Molecular Models

NGSS Aligned: Developing Models and Analyzing Computational Data

Name: _____ Date: _____ Class: _____

Activity Overview: In this lab, you will generate three distinct types of molecular models for two substances: Methane (CH_4) and Fluoromethane (CH_3F). Your goal is not just to see what the models look like, but to critically analyze how each model succeeds and where it fails to accurately represent the physical reality of a molecule.
Guiding Question: How do different molecular models help us visualize the physical and electronic forces that determine molecular structure?

Model 1: 2D Lewis Structures (Whiteboard)
Task 1: Draw the accurate 2D Lewis structure for both target molecules.

CH_4	*CH_3F*
What aspects of the Lewis Structure Model are clearly observable?	
A limitation of a model is something it *ignores* or misrepresents about the real world. What types of limitations does the Lewis Structure Model have?	

Model 2: 3D Physical Model (MolyMod Kits)

Task 2: Use your MolyMod kits to translate your Lewis structures into tactile, 3D shapes.

1. Obtain the necessary atoms and connecting bonds.
2. Assemble the molecules based *only* on the connectivity you drew for Task 1.
3. Ensure your final structure matches the common geometry types predicted by VSEPR.
4. Try to draw the MolyMod Structures as best as you can.

CH_4	CH_3F
What aspects of the 3D Physical (MolyMod) Model are clearly observable?	
A limitation of a model is something it *ignores* or misrepresents about the real world. What types of limitations does the 3D Physical *(MolyMod) Model* have?	

Model 3: 3D Computational Model (WebMO)

You will now use computational chemistry software (WebMO) to generate a scientifically optimized model of these molecules. This model will use real physics and mathematics to calculate realistic bond lengths and the actual distribution of charge.

Task 3: Access WebMO (webmo.net/demoserver/cgi-bin/webmo/login.cgi), on your device. When prompted for Username and Password, both are "guest." Complete the following sequence. Build Methane (CH_4) and Fluoromethane (CH_3F) in two separate tabs.

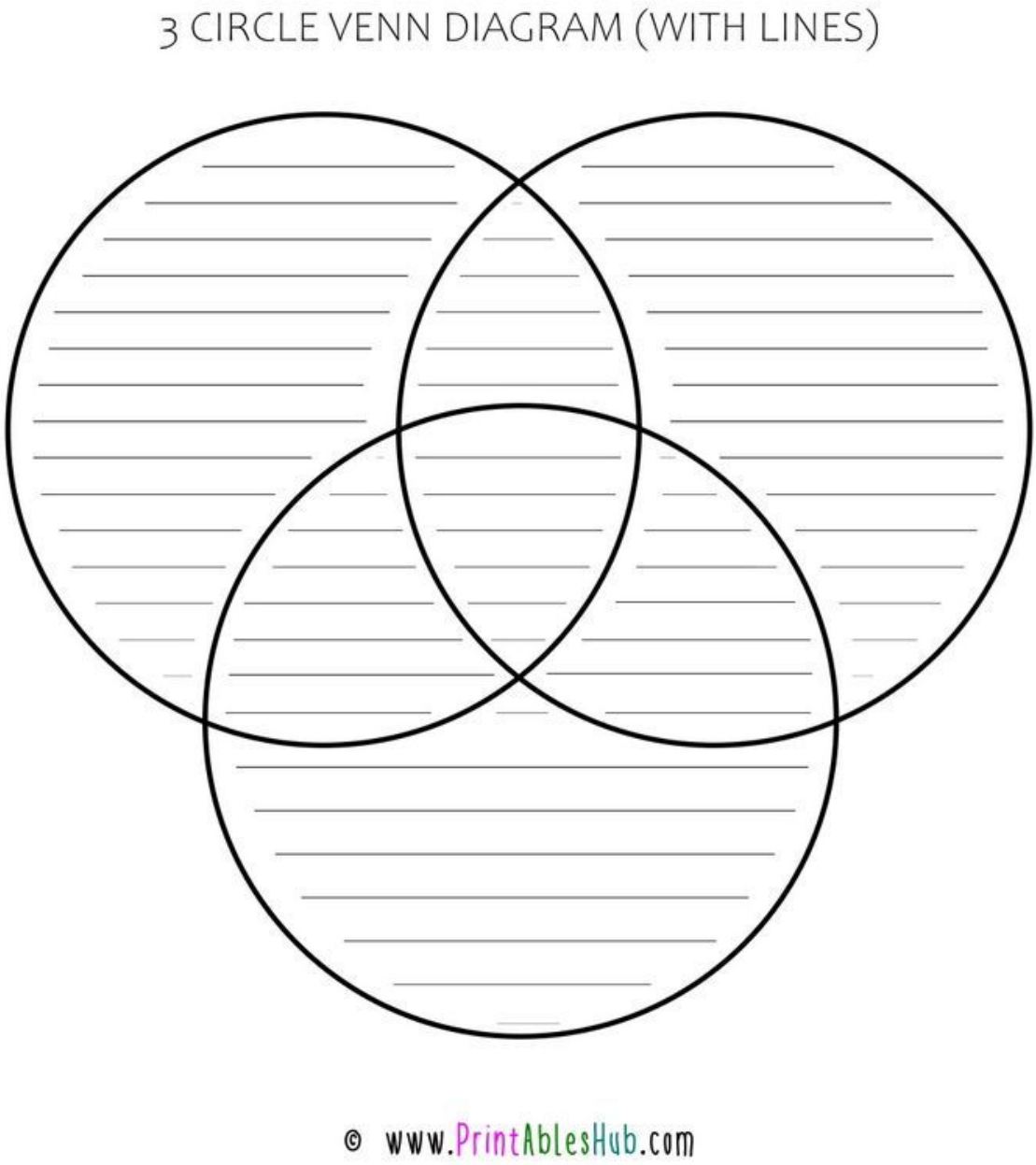
NOTE: You can hover the cursor over any of the icons in the left-side toolbar to see what they do.

1. **Create New Job:** New Job → Create New Job
2. **Draft:** Sketch the basic atomic connectivity using the Editor. (Note: It may look messy).
    a. On the left side toolbar, you can add new atoms by clicking on the Build Icon (looks like a water molecule).
    b. If you want to change the atom type, click on the Periodic Table Icon and select the type of atom you want to add.
    c. To make a bond between two atoms, click on one atom and drag the cursor to the other atom. Repeat if you want double or triple bonds.
3. **Optimize:** Go to Cleanup → Geometry. Observe the molecule snap into an optimized 3D shape, similar to your VSEPR shape.
4. **Bond length:** Click the Adjust Icon (looks like an arrow pointing upward left). Click two different atoms with a bond between them. The bond length is at the bottom left, measured in Angstroms. Make note of the bond lengths of C-H and C-F.
5. **Analyze Charge:** Go to Calculate → Huckel → Electrostatic Potential. Wait for the colorful gradient map to render on the molecular surface.
6. **Standardization of Charge Scale:**
    a. Options → Mapped Properties
    b. Change Min: -0.17 Max: 0.13
7. **Data Interpretation:**
    - **Red Color:** Electron-rich region $^{-}$, high potential for negative charge.
    - **Blue Color:** Electron-deficient region $^{+}$, high potential for positive charge.
8. Do your best to draw the images of the Electrostatic Potential Models using colored pencils as best as you can.

CH_4	CH_3F
What aspects of the 3D Computational (WebMO) Model are clearly observable?	
A limitation of a model is something it *ignores* or misrepresents about the real world. What types of limitations does the 3D Computational (WebMO) *Model* have?	

Comparison of Models Graphic Organizer:

Referring to your answers above, fill out the following Venn Diagram. Overlapping regions would be similar aspects represented by 2 or more Models.



Analysis Questions:

1. **Coulomb's Law:** Earlier in this class, we studied Coulomb's Law , which relates electrostatic force to charge and *distance*. Explain how atomic radii influence *bond length* and the attractive forces present in a chemical bond. *Hint: Consider what charges are attracting each other inside a bond.*

2. **Electronegativity:** Analyze your rendered ESP map for Fluoromethane (CH_3F). Where is the most intense RED region localized? Based on periodic trends, which atom is the most electronegative? Explain how high electronegativity *creates* that specific red region.

3. **Charge Distribution:** Compare the two maps. Does methane(CH_4) show significant charge separation? Why does the structure of methane (CH_4) create a "balanced" charge distribution, while fluoromethane (CH_3F) does not?

4. **Bond Polarity vs Molecular Polarity:** We know that bond polarity (separation of charge) is a result of the unequal sharing of electrons within a covalent between two atoms of significantly different electronegativity. Methane (CH_4) and fluoromethane (CH_3F) can be classified as being either a **polar molecule** or a **nonpolar molecule.** Utilize your ESP maps to support and clarify your reasoning.

CH_4 is a _____ molecule	CH_3F is a _____ molecule

5. **Independent Exploration:**
    - Utilize WebMO to construct a variety of molecular shapes. Begin by sketching a correct Lewis structure to verify that the target molecule can be successfully generated within WebMO. You are encouraged to experiment with different atomic elements and modify bond types, including double and triple bonds. Water would be a great molecule to create and explore!
    - Additionally, you can load established molecular structures, such as common over-the-counter pharmaceuticals (like acetaminophen or aspirin), by navigating to Lookup → Name on the upper toolbar. Try replacing specific atoms within these recognized compounds to observe the resulting impacts on the electrostatic potential map.
    - During this exploration period, what two discoveries did you make?